TEAM HERMIT CRAB · DELL X NVIDIA AI HACKATHON · NYC · AUG 22 2026

BEARING WITNESS.

DETECT THE CHANGE · EXPLAIN THE SPECTRUM
ESCALATE WITH EVIDENCE · A HUMAN HAS TO SAY YES

An always-on bearing-screening agent that runs where the data lives: entirely on the Dell Pro Max with GB10.

A REAL BEARING (SKF 129 HA) · PHOTO: WIKIMEDIA COMMONS, CC BY 4.0

41%

of all motor failures are bearings. The largest utility motor reliability study ever run: 6,312 motors.

ALBRECHT ET AL., IEEE TRANS. ENERGY CONVERSION, 1986

$125,000

median cost of ONE HOUR of unplanned downtime. Two-thirds of plants eat at least one a month.

ABB, VALUE OF RELIABILITY SURVEY, 2023 · N = 3,215 PLANTS

Most plants still check bearings once a month, by hand: their vibration data is not allowed to leave the building, and cloud AI cannot come in.

PHYSICS DOES THE DIAGNOSIS. A LOCAL MODEL FILES THE PAPERWORK. A HUMAN SAYS YES.

01 · DETECT

Persistent change against the asset's own early baseline.

02 · EXPLAIN

Known machine frequencies labelled before anything is called a fault.

03 · CORROBORATE

Two independent spectral views must agree, at slip-aware widths.

04 · ESCALATE

A typed work order with evidence locators. Approve, reject, or defer.

ALWAYS-ON · THE AGENT ACTS ON ITS OWN OVER TIME

CASES LAND WHILE NOBODY CLICKS.

- Watch mode replays the bearing's life through the engine on a cadence
- Every analysis writes to MongoDB through the same gated path
- Windows it cannot prove are **refused**, and the refusal is recorded
- Zero cloud calls. The box is the whole system.

THE FLEET · 15 OF 15 EVALUATED UNDER FROZEN V3 · EVERY CARD A REAL VERDICT

ONE CATCH INSIDE THE MONTHLY BLIND SPOT PAYS FOR THE BOX.

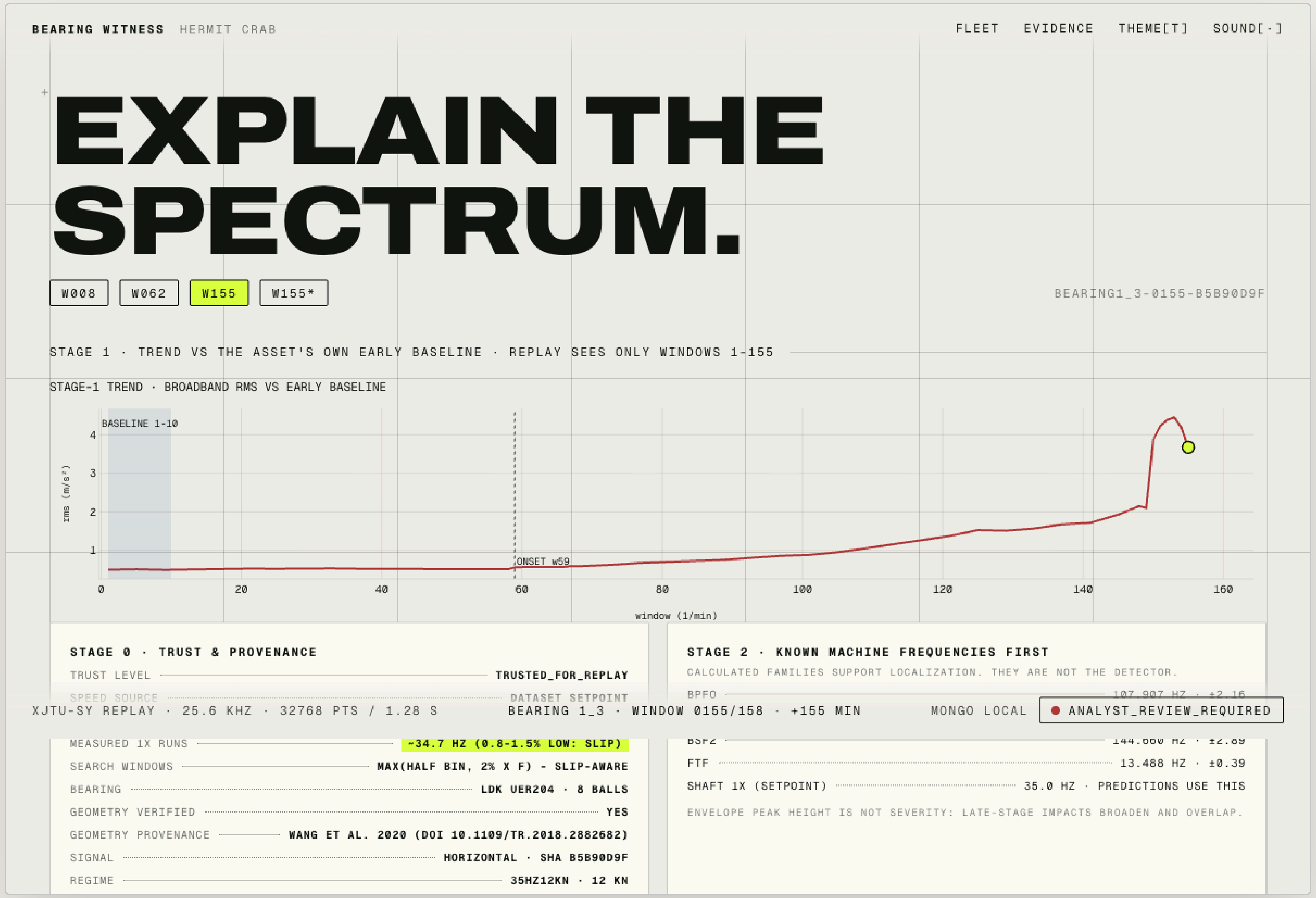
THE BUYER

Every OT-isolated plant that already refuses cloud monitoring: pharma, food, defense, chemicals, utilities. The customer who needs this most is exactly the one nothing else can serve.

THE PLANNER'S MORNING

Before: a monthly route, a clipboard, a guess.
After: a screening queue with evidence, and the work order already in the system of record.

AT ABB'S MEDIAN DOWNTIME RATE OF $125,000/HOUR (2023 SURVEY, N=3,215)



107.03 HZ. MEASURED.

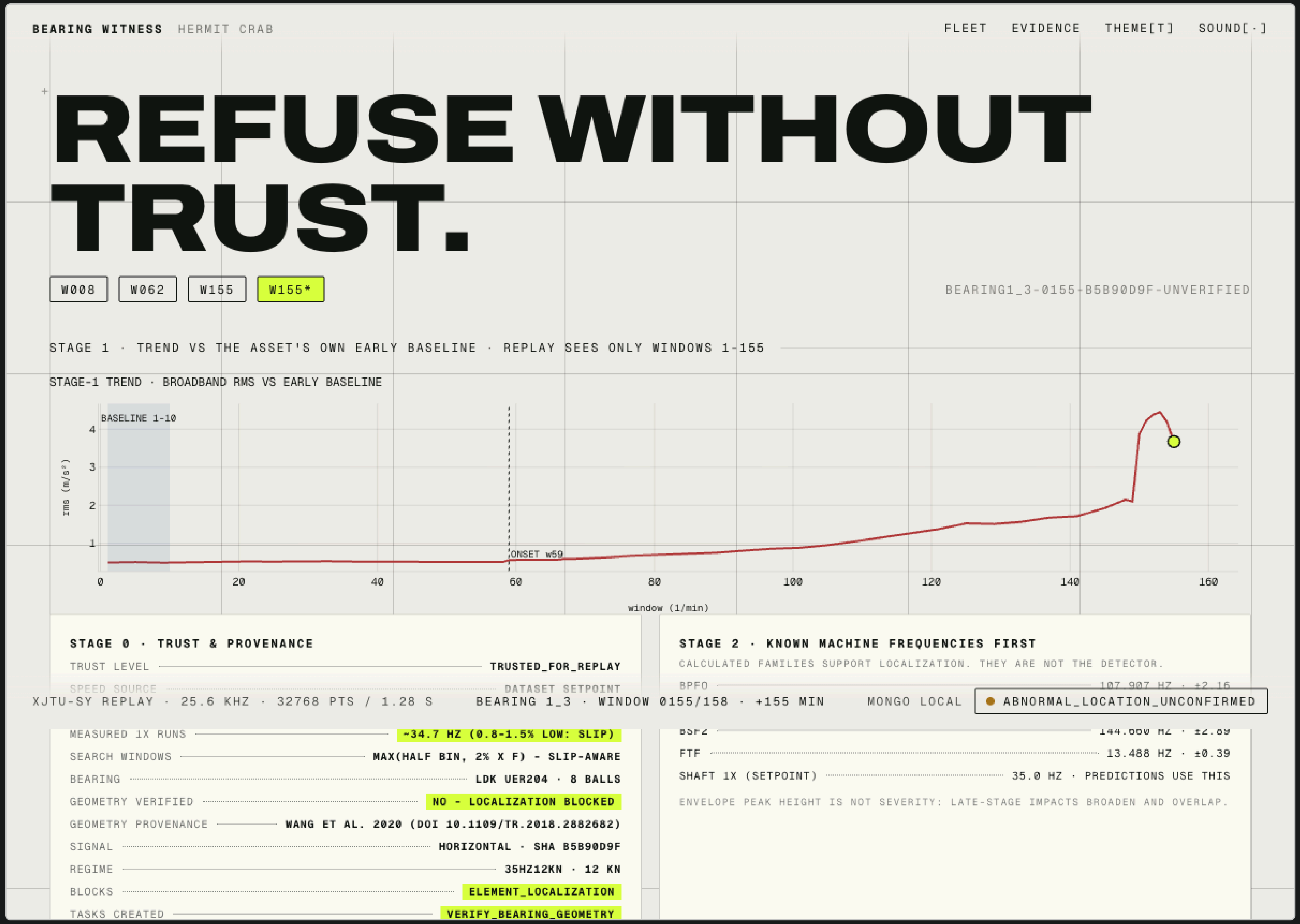
Prediction said 107.9. The measurement runs 0.8% low: real shaft slip, shown on screen, never hidden.

Two spectral views, three harmonics each, every peak carrying a clickable evidence locator with its file hash.

THERE IS NO APPROVE BUTTON.

Same waveform, geometry unverified: no location call, a verification task instead, and the UI cannot approve it. Neither can the database: the decision write is compare-and-set on the review state.

Most demos add AI. Ours constrains it.



GEOMETRY UNVERIFIED → LOCALIZATION REFUSED → VERIFY TASK DRAFTED

THE RECEIPTS · THRESHOLDS FROZEN BEFORE THE RUN · ALL 15 RUN-TO-FAILURE BEARINGS

0 / 0

WRONG CALLS / MISSED DETECTIONS

10 exact + 1 cage-consistent localizations. 4 honest abstentions, each carrying its machine reason. Inner and outer race faults both.

99 MIN

MEDIAN WARNING BEFORE END OF LIFE

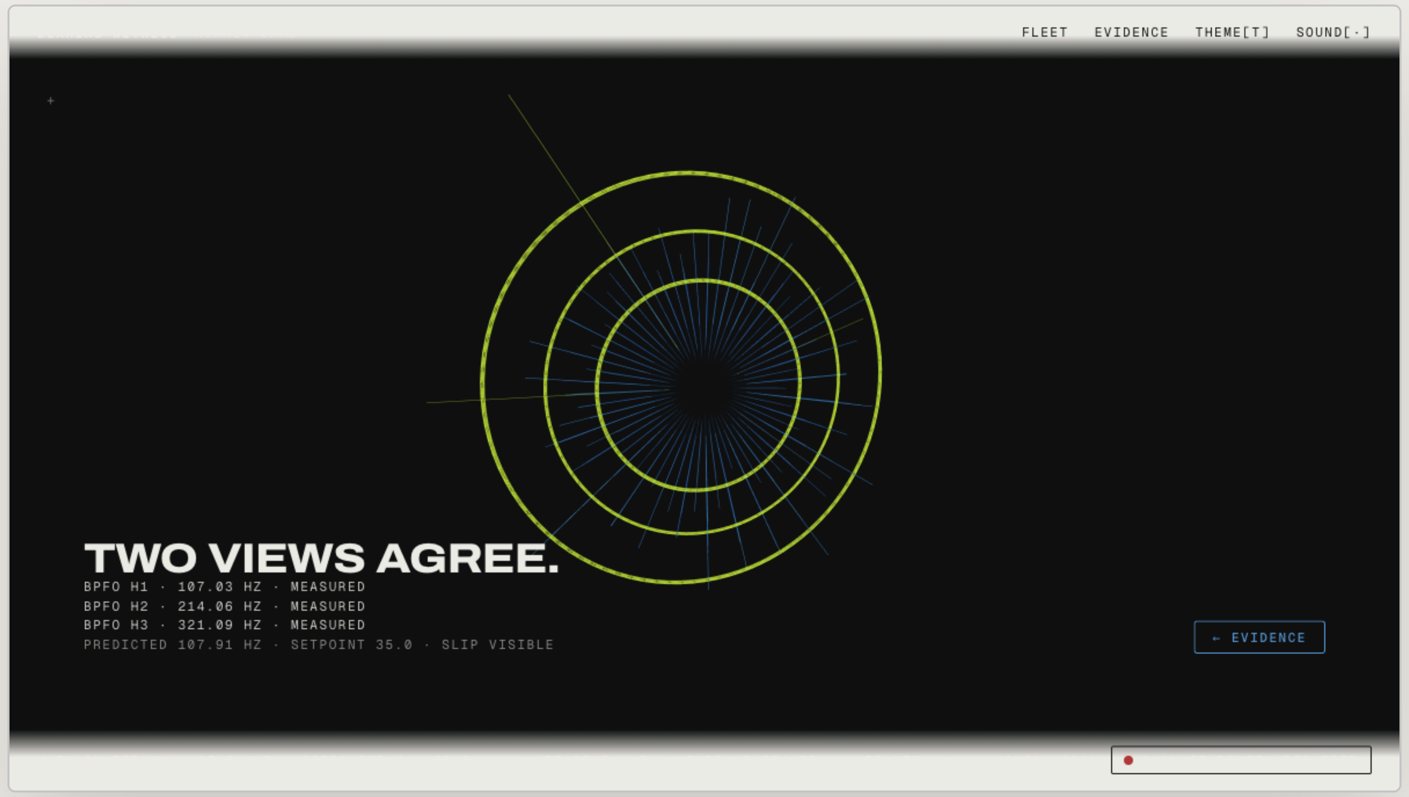
42 HOURS

BEST CASE (BEARING3_1, 2,519 MIN)

MEDIAN 98 EXCLUDING ONE STRUCTURAL FLOOR · WE STATE IT BOTH WAYS · FREEZE SHA IN THE COMMITTED OUTPUT

TECHNICAL EXECUTION · IT DOESN'T BREAK

- NVIDIA NemoClaw + OpenClaw + OpenShell (deny-by-default egress, shown live)
- Local Qwen3 27B (4-bit, Ollama) on the GB10 · MongoDB Community, localhost only
- SciPy owns diagnosis: deterministic, auditable, frozen thresholds
- 150+ tests · CI green · runs from a cold clone



THE CORROBORATION VIEW · WEBGL FED BY THE REAL ENVELOPE BINS

DETECT THE CHANGE. EXPLAIN THE SPECTRUM. ESCALATE WITH EVIDENCE. A HUMAN HAS TO SAY YES.

BUILT TODAY, ON THE BOX, ON REAL RUN-TO-FAILURE DATA · IT NEVER GUESSED ONCE

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